7

Inferences About the Difference Between Two Means

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Key Concepts

1. Independent versus dependent samples
2. Sampling distribution of the difference between two means
3. Standard error of the difference between two means
4. Parametric versus nonparametric tests

In Chapter 6 we introduced hypothesis testing and ultimately considered our first inferential statistic, the one-sample *t* test. There we examined the following general topics: types of hypotheses, types of decision errors, level of significance, steps in the decision-making process, inferences about a single mean when the population standard deviation is known (the *z* test), power, statistical versus practical significance, and inferences about a single mean when the population standard deviation is unknown (the *t* test).

In this chapter we consider inferential tests involving the difference between two means. In other words, our research question is the extent to which two sample means are statistically different and, by inference, the extent to which their respective population means are different. Several inferential tests are covered in this chapter, depending on whether the two samples are selected in an independent or dependent manner, and on whether the statistical assumptions are met. More specifically, the topics described include the following inferential tests: for two independent samples, the independent *t* test, the Welch *t′* test, and the Mann-Whitney-Wilcoxon test; for two dependent samples, the dependent *t* test and the Wilcoxon signed ranks test. We use many of the foundational concepts covered in Chapter 6. New concepts to be discussed include the following: independent versus dependent samples; the sampling distribution of the difference between two means; and the standard error of the difference between two means. Our objectives are that by the end of this chapter, you will be able to: (a) understand the basic concepts underlying the inferential tests of two means, (b) select the appropriate test, and (c) determine and interpret the results from the appropriate test.

7.1 Inferences About Two Independent Means and How They Work

Remember our very capable quad of graduate students who work in the stats lab? Let's see what Oso Wyse and Addie Venture have in store now . . .

The stats lab has been humming with research project requests from faculty and the community. The latest request comes from Dr. Nightingale, a local nurse practitioner, who is studying cholesterol levels of adults and how they differ based on sex. Oso Wyse has been assigned to the project and suggests the following research question: *Is there a mean difference in cholesterol level between males and females?* Oso suggests an independent samples *t* test as the test of inference. His task is then to assist Dr. Nightingale in generating the test of inference to answer the research question.

Addie Venture has been asked to consult with the institution's swimming coach, Coach Bryant, who works with the community and various swimming programs that are offered through their local Parks & Recreation Department. Coach Bryant has just

categories.) It is *not* a condition of the independent *t* test that the sample sizes of the two groups be the same. *An unbalanced design (i.e., unequal sample sizes) is perfectly acceptable*. An unbalanced design is only a concern in the event that the assumption of homogeneity is violated. If you find yourself in that situation, please refer to the previous discussion on measures that can be taken.

7.2 Inferences About Two Dependent Means and How They Work

In this section, two inferential tests of the difference between two dependent means are described, the dependent *t* test and briefly the Wilcoxon signed ranks test. The section concludes with a list of recommendations.

7.2.1 Characteristics of the Dependent *t* Test

As you may recall, the **dependent *t* test** is appropriate to use when there are two samples that are dependent; that is, the individuals in sample 1 have some relationship to the individuals in sample 2. Although there are several methods for computing the test statistic *t*, the most direct method and the one most closely aligned conceptually with the one-sample *t* test is as follows:

$$t = \frac{\bar{d}}{s_{\bar{d}}}$$

where $\bar{d}$ is the **mean difference**, and $s_{\bar{d}}$ is the **standard error of the mean difference**. Conceptually, this test statistic looks just like the one-sample *t* test statistic, except now the notation has been changed to denote that we are dealing with *difference scores* rather than raw scores.

The **standard error of the mean difference** is computed by

$$s_{\bar{d}} = \frac{s_d}{\sqrt{n}}$$

where s_d is the standard deviation of the difference scores (i.e., like any other standard deviation, only this one is computed from the difference scores rather than raw scores), and n is the total number of pairs. Conceptually, this standard error looks just like the standard error for the one-sample *t* test. If we were doing hand computations, we would compute a difference score for each pair of scores (i.e., $Y_1 - Y_2$). For example, if sample 1 were wives and sample 2 were their husbands, then we calculate a difference score for each couple. From this set of difference scores, we then compute the mean of the difference scores $\bar{d}$ and standard deviation of the difference scores, s_d. This leads us directly into the computation of the *t* test statistic. Note that although there are n scores in sample 1, n scores in sample 2, and thus $2n$ total scores, there are only n difference scores, which is what the analysis is actually based upon.

7.3 Computing Inferences About Two Independent Means Using SPSS

Instructions for determining the independent samples *t* test using SPSS are presented first. The data-screening section provides additional steps for examining the assumption of normality for the independent *t* test.

Step 1. In order to conduct an independent *t* test, your dataset needs to include one dependent variable *Y* that is measured on an interval or ratio scale (e.g., "cholesterol") as well as a grouping variable *X* that is measured on a nominal or ordinal scale (e.g., "gender"). For the grouping variable, if there are more than two categories available, only two categories can be selected (or multiple categories must be collapsed so there are only two categories) when running the independent *t* test (the analysis of variance is required for examining more than two categories). To conduct the independent *t* test, go to the "Analyze" in the top pulldown menu, select "Compare Means," and then select "Independent-Samples T Test." Following the steps in the screenshot in Figure 7.5 produces the "Independent-Samples T Test" dialog box.

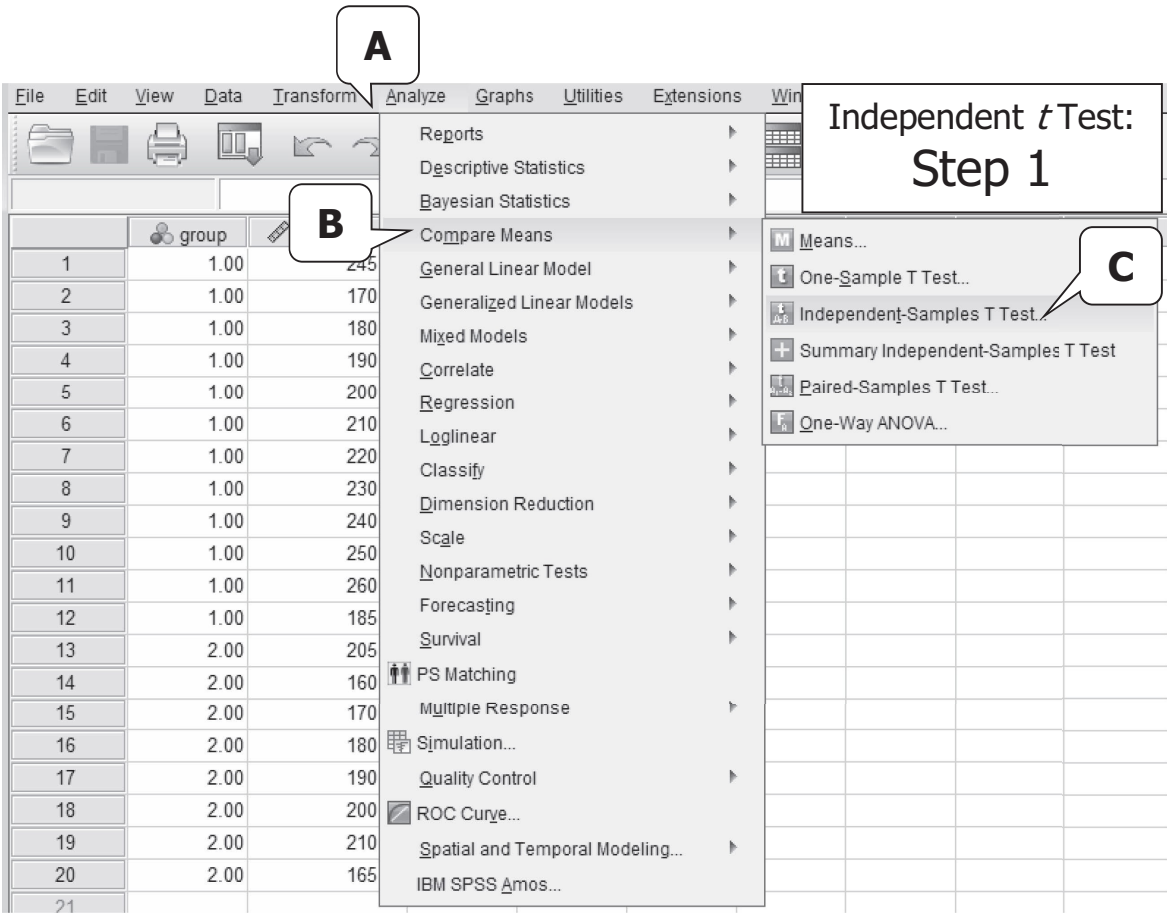
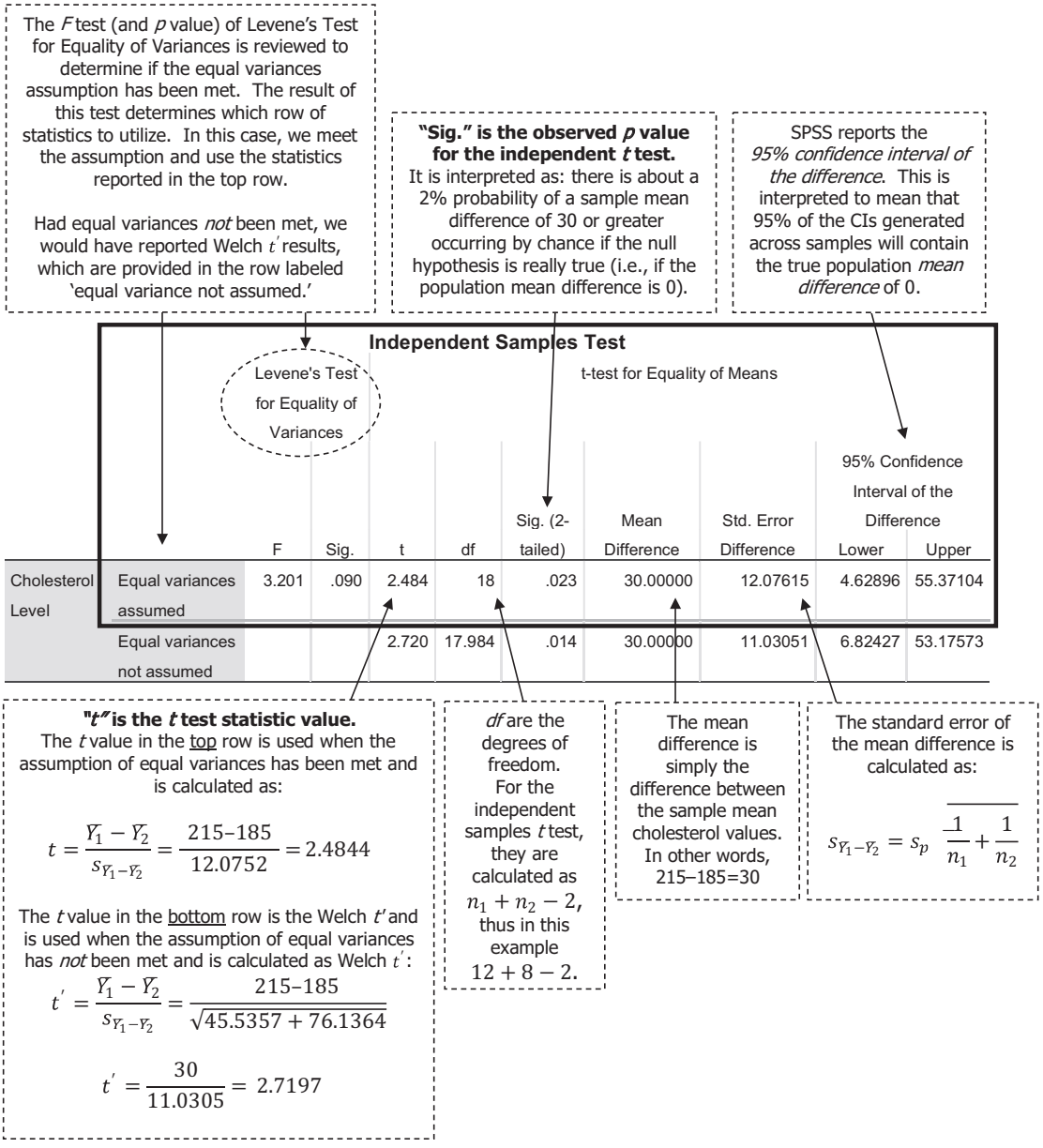


FIGURE 7.5
Independent *t* test: Step 1.

Step 2. Next, from the main "Independent-Samples T Test" dialog box, click the dependent variable (e.g., "cholesterol") and move it into the "Test Variable" box by clicking the arrow button. Next, click the grouping variable (e.g., "gender") and move it into the "Grouping

TABLE 7.5 (continued)

SPSS Results for Independent *t* Test



the observed probability, p, is less than our nominal alpha of .05, we reject the null hypothesis. There is a statistically significant difference between groups, and there is evidence to suggest that the mean cholesterol level for males is different than the mean cholesterol level for females.

7.4 Computing Inferences About Two Dependent Means Using SPSS

Next, instructions for determining the dependent samples *t* test using SPSS are presented. The data-screening section provides additional steps for examining the assumptions of normality and homogeneity for the dependent *t* test.

7.5 Computing Inferences About Two Independent Means Using R

Next we consider **R** for the dependent *t* test. The scripts are provided within the blocks with additional annotation to assist in understanding how the commands work. Should you want to write reminder notes and annotation to yourself as you write the commands in **R** (and we highly encourage doing so), remember that any text that follows a hashtag (i.e., #) is annotation only and not part of the **R** script. Thus, you can write annotations directly into **R** with hashtags. We encourage this practice so that when you call up the commands in the future, you'll understand what the various lines of code are doing. You may think you'll remember what you did. However, trust us. There is a good chance that you won't. Thus, consider it best practice when using **R** to annotate heavily!

7.5.1 Reading Data into R

```
getwd()
```

R is always pointed to a directory on your computer. To find out which directory it is pointed to, run this "get working directory" function. We will assume that we need to change the working directory, and will use the next line of code to set the working directory to the desired path.

```
setwd("E:/Folder")
```

To set the working directory, use the *setwd* function and change what is in quotation marks here to your file location. Also, if you are copying the directory name, it will copy in slashes. You will need to change the slash (i.e., \) to a forward slash (i.e., /). Note that you need your destination name within quotation marks in the parentheses.

```
Ch7_cholesterol <- read.csv("Ch7_cholesterol.csv")
```

The *read.csv* function reads our data into **R**. What's to the left of the <- will be what the data will be called in **R**. In this example, we're calling the **R** dataframe "Ch7_cholesterol." What's to the right of the <- tells **R** to find this particular .csv file. In this example, our file is called "Ch7_cholesterol.csv." Make sure the extension (i.e., .csv) is included in your script. Also note that the name of your file should be in quotation marks within the parentheses.

```
names(Ch7_cholesterol)
```

The *names* function will produce a list of variable names for each dataframe as follows. This is a good check to make sure your data have been read in correctly.

```
[1] "group" "cholesterol"
```

```
View(Ch7_cholesterol)
```

The *View* function will let you view the dataset in spreadsheet format in RStudio.

FIGURE 7.10
Reading data into **R**.

```
Odds Ratio ES:

OR [ 95 %CI] = 7.82 [ 1.21 , 50.67 ]
  p-value(OR) = 0.03

Log OR [ 95 %CI] = 2.06 [ 0.19 , 3.93 ]
  var(lOR) = 0.79
  p-value(Log OR) = 0.03

Other:

NNT = 2.41
Total N = 20
```

FIGURE 7.11 (continued)
Generating the independent *t* and Welch *t'* tests.

7.6 Computing Inferences About Two Dependent Means Using R

Next we consider **R** for the dependent *t* test. As noted previously, the scripts are provided within the blocks with additional annotation to assist in understanding how the commands work.

7.6.1 Reading Data Into R

```
getwd()
```

R is always pointed to a directory on your computer. To find out which directory it is pointed to, run this "get working directory" function. We will assume that we need to change the working directory, and will use the next line of code to set the working directory to the desired path.

```
setwd("E:/Folder")
```

To set the working directory, use the *setwd* function and change what is in quotation marks here to your file location. Also, if you are copying the directory name, it will copy in slashes. You will need to change the slash (i.e., \) to a forward slash (i.e., /). Note that you need your destination name within quotation marks in the parentheses.

```
Ch7_swim <- read.csv("Ch7_swim.csv")
```

The *read.csv* function reads our data into **R**. What's to the left of the <- will be what the data will be called in **R**. In this example, we're calling the **R** dataframe "Ch7_swim." What's to the right of the <- tells **R** to find this particular .csv file. In this example, our file is called "Ch7_swim.csv." Make sure the extension (i.e., .csv) is included in your script. Also note that the name of your file should be in quotation marks within the parentheses.

FIGURE 7.12
Reading data into **R** for the dependent *t* test.

```
Ch7_depT
```

This script will generate output from the test we just conducted. We see that our test statistic is 7.3193, with 9 degrees of freedom, and p value of < .001. The 95% confidence interval of the mean difference is 3.45 to 6.54. The mean of the differences is 5.

```
        Paired t-test

data: Ch7_swim$pretest and Ch7_swim$posttest

t = 7.3193, df = 9, p-value = 4.472e-05

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:
 3.454652 6.545348

sample estimates:
mean of the differences
                      5
```

FIGURE 7.13 (continued)
Generating the dependent *t* test.

7.7 Data Screening

We will begin data screening with examining the extent to which the assumptions of the independent *t* test were met. This will be following by data screening for the assumptions of the dependent *t* test.

7.7.1 Data Screening for the Independent *t* Test

The assumptions for the independent *t* test that we need to examine via data screening include the *normality* of the distribution of the dependent variable by categories of the independent variable and *homogeneity of variances*. Recall that the assumption of independence is required as well; however, as noted earlier, that is not an assumption with which data will be used to assess the extent to which the assumption is met.

7.7.1.1 Normality for the Independent t *Test*

Let's first examine the assumption of normality of the distribution of the dependent variable by categories of the independent variable. As alluded to earlier in the chapter, understanding the distributional shape, specifically the extent to which normality is a reasonable assumption, is important. *For the independent* t *test, the distributional shape for the dependent variable should be normally distributed for each category/group of the independent variable*. As with our one-sample *t* test, we can again use Explore to examine the extent to which the assumption of normality is met.

The general steps for accessing Explore have been presented in previous chapters (e.g., Chapter 4), and they will not be reiterated here. Normality of the dependent variable

7.7.2.2 Homogeneity of Variance for the Dependent t *Test*

We also need to examine evidence for meeting equal variances, or more specifically homogeneity of variance of the difference scores. Without conducting a formal test of equality of variances (as we do in Chapter 9), a rough benchmark for having met the assumption of homogeneity of variances when conducting the dependent *t* test is that the *ratio of the smallest to largest variance of the paired samples is no greater than 1:4 to decrease the chance of a Type I error. Recent research suggests that a variance ratio lower than 1.5 should be used as convention in the presence of heterogeneity with unequal sample sizes* (Blanca, Alarcón, Arnau, Bono, & Bendayan, 2018). The variance can be computed easily by any number of procedures in SPSS (refer back to Chapter 3, for example), and these steps will not be repeated here. For our paired samples, the variance of the pretest score is 17.778 and the variance of the posttest score is 13.111—well within the range of 1:4 suggesting that homogeneity of variances is reasonable.

7.8 G*Power

Using the results of the independent samples *t* test just conducted, let's use G*Power to compute the post hoc power of our test.

7.8.1 Post Hoc Power for the Independent *t* Test Using G*Power

The first thing that must be done when using G*Power for computing post hoc power is to select the correct test family. In our case, we conducted an independent samples *t* test, therefore the default selection of "t tests" is the correct test family. Next, we need to select the appropriate statistical test. We use the arrow to toggle to "Means: Difference between two independent means (two groups)." The "Type of power analysis" then needs to be selected. To compute post hoc power, we need to select "Post hoc: Compute achieved power—given α, sample size, and effect size."

The "Input Parameters" must then be specified. The first parameter is the selection of whether the test is one tailed (i.e., directional) or two tailed (i.e., nondirectional). In this example, we have a two-tailed test, so we use the arrow to toggle to "Two." We can input our observed effect size, $d_{\text{¸}}$ or we can also use the pop-out calculator to compute the effect size *d*. Using the pop-out calculator, our effect size is 1.19 (note that the pop-out calculator does not use the pooled standard deviation as the standardizer). The alpha level we tested at was .05, and the sample size for females was 8 and for males, 12. Once the parameters are specified, simply click "Calculate" to generate the achieved power statistics.

The "Output Parameters" provide the relevant statistics given the input just specified. In this example, we were interested in determining post hoc power given a two-tailed test, with an observed effect size of 1.18, an alpha level of .05, and sample sizes of 8 (females) and 12 (males). Based on those criteria, the post hoc power was .69. In other words, with a sample size of 8 female and 12 males in our study, testing at an alpha level of .05 and observing a large effect of 1.19, then the power of our test was .69—the probability of rejecting the null hypothesis when it is really false will be 69%, which is only moderate power (minimally acceptable power is generally about 80%). Keep in mind that conducting power analysis *a priori* is recommended so that you avoid a situation where, post hoc, you find that the sample size was not sufficient to reach the desired power (given the

The "Input Parameters" must then be specified. The first parameter is the selection of whether your test is one tailed (i.e., directional) or two tailed (i.e., nondirectional). In this example, we have a two-tailed test, so we use the arrow to toggle to "Two." The achieved or observed effect size was 2.3146. The alpha level we tested at was .05, and the total sample size was 10. Once the parameters are specified, simply click "Calculate" to generate the achieved power statistics.

The "Output Parameters" provide the relevant statistics given the input specified. In this example, we were interested in determining post hoc power given a two-tailed test, with an observed effect size of 2.3146, an alpha level of .05, and total sample size of 10. Based on those criteria, the post hoc power was .99. In other words, with a total sample size of 10, testing at an alpha level of .05 and observing a large effect of 2.3146, then the power of our test was greater than .99—the probability of rejecting the null hypothesis when it is really false will be greater than 99%, about the strongest power that can be achieved. Again, conducting power analysis *a priori* is recommended so that you avoid a situation where, post hoc, you find that the sample size was not sufficient to reach the desired power (given the observed effect size and alpha level).

7.9 Research Question Template and Example Write-Up

Next we develop APA-style paragraphs describing the results for both examples. First is a paragraph describing the results of the independent *t* test for the cholesterol example, followed by dependent *t* test for the swimming example.

7.9.1 Research Question Template and Example Write-Up for the Independent *t* Test

Recall that our graduate research assistant, Oso, was working with Dr. Nightingale, a local nurse practitioner, to assist in analyzing cholesterol levels. His task was to assist Dr. Nightingale with writing her research question (*Is there a mean difference in cholesterol level between males and females?*) and generating the test of inference to answer her question. Oso suggested an independent-samples *t* test as the test of inference. A template for writing a research question for an independent *t* test follows:

> Is there a mean difference in [dependent variable] between [group 1 of the independent variable] and [group 2 of the independent variable]?

It may be helpful to preface the results of the independent-samples *t* test with information on an examination of the extent to which the assumptions were met (recall there are three assumptions: normality, homogeneity of variances, and independence). This assists the reader in understanding that you were thorough in data screening prior to conducting the test of inference.

> An independent samples *t* test was conducted to determine if the mean cholesterol level of males differed from females. The assumption of normality was tested and met

7.10 Additional Resources

A number of resources are available for learning more about statistics and how to interpret statistics. In addition to those already cited, Huck (2000) is an excellent general resource to assist in learning more about statistics and how to interpret statistics.

Problems

Conceptual Problems

1. When H_0 is true, the difference between two independent sample means is a function of which of the following?

a. Degrees of freedom
b. Standard error
c. Sampling distribution
d. Sampling error

2. The denominator of the independent *t* test is known as the standard error of the difference between two means, and may be defined as which of the following?

a. The difference between the two group means
b. The amount by which the difference between the two group means differs from the population mean
c. The standard deviation of the sampling distribution of the difference between two means
d. All of the above
e. None of the above

3. In the independent *t* test, what does the homoscedasticity assumption state?

a. The two population means are equal
b. The two population variances are equal
c. The two sample means are equal
d. The two sample variances are equal

4. True or false? Sampling error increases with larger samples.

5. True or false? At a given level of significance, it is possible that the significance test and the confidence interval results will differ for the same dataset.

6. I assert that the critical value of *t* required for statistical significance is smaller (in absolute value) when using a directional rather than a nondirectional test. Am I correct?

7. If a 95% CI from an independent *t* test ranges from –.13 to +1.67, I assert that the null hypothesis would not be rejected at the .05 level of significance. Am I correct?

8. The mathematic ability of 10 preschool children was measured when they entered their first year of preschool and then again in the spring of their kindergarten year. To test for pre to post mean differences, which of the following tests would be used?